Purity of Pure and Recombination

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We prove that, over the background theory T_0, Purity of Pure at type $t \to t$ is inconsistent with unary Recombination and the existence of a fundamental proposition. We also prove that T_0 and this single instance of Purity of Pure rule out a fun′ proposition.

We use Goodman's notation and put $u = t \to t$. Lower-case variables p, q, r, s range over propositions. Upper-case variables V, W, X, Y, B range over unary operators of type u. The symbol $\mathcal{Z}$ ranges over properties of operators, of type $u \to t$. We write

$$\Box A = (A = \top), \qquad V \subseteq W \quad \text{for} \quad \forall p\,(Vp \to Wp).$$

The inclusion $V \subseteq W$ says that every proposition satisfying V satisfies W. When we need necessary inclusion, we write $\Box(V \subseteq W)$.

The background theory T_0 includes Classicism, the Purity schema and Application closure. The Purity schema says that every closed logical term is pure. Application closure is the schema

$$\mathrm{Pure}_{\sigma\to\tau}(F)\,\&\,\mathrm{Pure}_{\sigma}(a) \to \mathrm{Pure}_{\tau}(Fa).$$

Let $\mathcal{T}$ be the axiom extension of T_0 by

(PP_u) $$\mathrm{Pure}_{u\to t}(\mathrm{Pure}_u).$$

The Rule of Equivalence applies to theorems of this extension. Consequently, if $\mathcal{T} \vdash A$, then $\mathcal{T} \vdash A = \top$, and thus $\mathcal{T} \vdash \Box A$. We use this rule of Necessitation, the S4 laws, Boolean identities, β-conversion and typed universal instantiation. In Lemma 6 we explicitly mark the application of Necessitation to a theorem whose proof uses (PP_u).

Theorem 1. *The theory $\mathcal{T}$ proves that no proposition r satisfies*

(R) $$\forall X^u\big(\mathrm{Pure}(X) \to (\Box(Xr) \to \forall p\, Xp)\big).$$

Hence $\mathcal{T} \vdash {\sim}\exists r\,\mathrm{fun}'(r)$, and

$$\mathcal{T} + \textit{unary Recombination} \vdash {\sim}\exists r\,\mathrm{Fun}(r).$$

Unary Recombination requires (R) of every fundamental proposition. The definition of $\mathrm{fun}'(r)$ is

$$\forall X^u Y^u \big((\mathrm{Pure}(X) \,\&\, \mathrm{Pure}(Y)) \to (Xr = Yr \to X = Y)\big).$$

We prove the theorem by constructing two unary operators, L and P. The closed logical operator L has the modal properties needed for the diagonal argument. The operator P collects the instances of a definable class of pure operators called stages. PP makes P pure and allows us to prove $L \subseteq P$. For each proposed r, we then obtain a proposition $q = r \,\&\, Lr$ and a pure stage Y_0 that omits q but whose successor contains it. This produces a pure operator X with $\Box(Xr)$ for which $X(\sim q)$ is false, contradicting (R).

A logical fixed-point construction. Define

$$\Sigma = \lambda Y^u.\lambda p.\big(\Box(p \vee Yp \vee Y(\sim p)) \vee \Box(\sim p \vee Yp \vee Y(\sim p))\big), \tag{1}$$

$$\mathrm{Pre}(W) = \Box(\Sigma W \subseteq W), \tag{2}$$

$$L = \lambda p.\forall W^u\,(\mathrm{Pre}(W) \to Wp). \tag{3}$$

Here Σ has type $u \to u$, Pre has type $u \to t$, and L has type u. Each abbreviation expands to a closed logical term. The condition $\mathrm{Pre}(W)$ says that W necessarily contains its Σ-successor. The operator L applies to precisely the propositions satisfying every such W.

Lemma 1. *The following are logical theorems.*

$$(\Sigma Y)(\sim p) \leftrightarrow (\Sigma Y)p, \tag{4}$$

$$(\Sigma Y)p \to \Box((\Sigma Y)p), \tag{5}$$

$$\Box(V \subseteq W) \to \Box(\Sigma V \subseteq \Sigma W). \tag{6}$$

Proof. Replacing p by $\sim p$ in (1) interchanges the two boxed disjuncts and interchanges Yp with $Y(\sim p)$ inside each box. Double negation and commutativity of disjunction give (4). For (5), S4 proves

$$\Box A \vee \Box B \to \Box(\Box A \vee \Box B).$$

Apply this to the two disjuncts in (1).

For monotonicity, suppose that $\Box(V \subseteq W)$. Boxed universal instantiation gives both $\Box(Vp \to Wp)$ and $\Box(V(\sim p) \to W(\sim p))$. The normal modal rules therefore take each disjunct of $(\Sigma V)p$ to its corresponding disjunct in $(\Sigma W)p$. Generalizing p and discharging the assumption proves

$$\Box(V \subseteq W) \to (\Sigma V \subseteq \Sigma W). \tag{7}$$

Necessitate the universal closure of this theorem and instantiate inside the box at V, W. Axiom 4 gives $\Box(V \subseteq W) \to \Box\Box(V \subseteq W)$, so K gives the boxed conclusion in (6). ∎

Lemma 2. *Logic alone proves*

$$Lp \leftrightarrow (\Sigma L)p, \tag{8}$$

$$Lp \to \Box(Lp), \tag{9}$$

$$L(\sim p) \leftrightarrow Lp. \tag{10}$$

It also proves $\mathrm{Pre}(W) \to \Box(L \subseteq W)$ *for every* W.

Proof. Instantiating the quantifier in (3) gives the logical theorem $\mathrm{Pre}(W) \to L \subseteq W$. Its necessitation, together with $\mathrm{Pre}(W) \to \Box\mathrm{Pre}(W)$ from axiom 4, gives

$$\mathrm{Pre}(W) \to \Box(L \subseteq W). \tag{11}$$

We first show $\Sigma L \subseteq L$. Suppose that $(\Sigma L)p$ and let W satisfy $\mathrm{Pre}(W)$. By (11) and (7), $(\Sigma W)p$. Factivity applied to $\mathrm{Pre}(W)$ then gives Wp. Discharging $\mathrm{Pre}(W)$ and generalizing W gives Lp by (3). After discharging $(\Sigma L)p$ and generalizing p, we have proved $\Sigma L \subseteq L$ as a logical theorem. Necessitation therefore gives

$$\Box(\Sigma L \subseteq L),$$

which is $\mathrm{Pre}(L)$.

Now apply the stronger monotonicity law (6) with $V = \Sigma L$ and $W = L$. It gives

$$\Box(\Sigma(\Sigma L) \subseteq \Sigma L),$$

which is $\mathrm{Pre}(\Sigma L)$. Instantiating (3) at $W = \Sigma L$ thus proves $L \subseteq \Sigma L$. Together the two inclusions give (8).

The fixed-point biconditional is a logical theorem, so its necessitation is available. From Lp, it gives $(\Sigma L)p$, then $\Box((\Sigma L)p)$ by (5), and finally $\Box(Lp)$ by K and the necessitated reverse implication. This proves (9). The symmetry law (10) follows by applying (8) on both sides of (4) with $Y = L$. ∎

Lemma 3. *Put $g = \lambda p.(p \,\&\, Lp)$. For each proposition r, put $q = gr$. Then logic proves Lq and $g(\sim q) = \sim q$.*

Proof. Conjoining Lr to r gives a proposition q for which we can first prove $q \to Lq$. Assume q, so Lr. By (9), $\Box(Lr)$, which by definition is $Lr = \top$. Hence

$$q = r \,\&\, Lr = r \,\&\, \top = r.$$

Substitution in Lr gives Lq. Discharging q and generalizing r proves

$$\forall r\big((r \,\&\, Lr) \to L(r \,\&\, Lr)\big).$$

Necessitation and boxed universal instantiation therefore give $\Box(\sim q \vee Lq)$. By weakening inside the box we obtain

$$\Box(\sim q \vee Lq \vee L(\sim q)).$$

This is the second disjunct of $(\Sigma L)q$, so (8) gives Lq. Symmetry gives $L(\sim q)$, and (9) gives $L(\sim q) = \top$. Thus

$$g(\sim q) = \sim q \,\&\, L(\sim q) = \sim q \,\&\, \top = \sim q. \qquad \blacksquare$$

The identities $gr = q$ and $g(\sim q) = \sim q$ will determine the values of our final operators at the two arguments used in the contradiction. Both L and g are closed logical operators and are therefore pure by the Purity schema.

The pure stages. Let $K_\perp = \lambda p.\perp$. A property $\mathcal{Z}$ of operators is called *closed* when it satisfies the following conditions.

1. $\mathcal{Z}(K_\perp)$.
2. For every pure Y, if $\mathcal{Z}(Y)$, then $\mathcal{Z}(\Sigma Y)$.
3. For every pure B, if each proposition s with Bs has a pure operator Y satisfying $\mathcal{Z}(Y)$, $Y \subseteq B$, and Ys, then $\mathcal{Z}(B)$.

The third condition admits a pure B whenever its extension is covered by the extensions of pure members of $\mathcal{Z}$ that it contains. We call this the union condition. Its formal expression is

$$\forall B\Big(\big(\mathrm{Pure}(B) \,\&\, \forall s\big(Bs \to \exists Y(\mathrm{Pure}(Y) \,\&\, \mathcal{Z}(Y) \,\&\, Y \subseteq B \,\&\, Ys)\big)\big) \to \mathcal{Z}(B)\Big).$$

Let $\mathrm{Cl}(\mathcal{Z})$ be the conjunction of the three conditions, and define

$$\mathrm{Stage}(Y) = \mathrm{Pure}(Y) \,\&\, \forall \mathcal{Z}\,(\mathrm{Cl}(\mathcal{Z}) \to \mathcal{Z}(Y)), \tag{12}$$

$$P = \lambda p.\exists Y\,(\mathrm{Stage}(Y) \,\&\, Yp). \tag{13}$$

Here Cl has type $(u \to t) \to t$, Stage has type $u \to t$, and P has type u. The quantifier over $\mathcal{Z}$ ranges over properties of type $u \to t$. The first conjunct in (12) makes every stage pure. Equation (13) gives $Y \subseteq P$ for every stage Y.

Lemma 4. *The property* Stage *is closed. If a property* Ψ *of type* $u \to t$ *satisfies the following three conditions, then every stage satisfies* Ψ*.*

1. $\Psi(K_\perp)$*.*
2. *If* Y *is a stage satisfying* Ψ*, then* $\Psi(\Sigma Y)$*.*
3. *If* B *is pure and each* s *with* Bs *has a stage* $Y \subseteq B$ *with* $\Psi(Y)$ *and* Ys*, then* $\Psi(B)$*.*

Parameters are allowed in Ψ*. Bound variables are chosen fresh for them.*

Proof. The operator $K_\perp$ is pure by the Purity schema and satisfies every closed property by its first condition. Hence it is a stage. If Y is a stage, then ΣY is pure by logical purity of Σ and Application closure. Every closed property contains Y and hence ΣY, making ΣY a stage.

For the union condition, suppose that B is pure and every s with Bs has a stage $Y \subseteq B$ with Ys. Fix a closed property $\mathcal{Z}$. Each such Y is pure and satisfies $\mathcal{Z}$, by (12). The third condition for $\mathcal{Z}$ gives $\mathcal{Z}(B)$. Generalizing $\mathcal{Z}$ proves that B is a stage. Thus Stage is closed.

Now suppose that Ψ satisfies the displayed conditions and put

$$\mathcal{W} = \lambda Y^u.(\mathrm{Stage}(Y) \,\&\, \Psi(Y)).$$

The first and second closure conditions for $\mathcal{W}$ follow from those for Stage and the corresponding conditions on Ψ. For the third, the witnesses in $\mathcal{W}$ are stages satisfying Ψ, so the union condition for Stage gives $\mathrm{Stage}(B)$ and the third condition on Ψ gives $\Psi(B)$. Thus $\mathrm{Cl}(\mathcal{W})$. Instantiating (12) at this typed λ-term shows that every stage satisfies Ψ. ∎

Lemma 5. *Every stage* Y *satisfies*

$$\forall p\, (Y(\sim p) \leftrightarrow Yp). \tag{14}$$

If a stage Y *satisfies* Yp*, then there is a stage* Y_0 *such that*

$$\sim Y_0 p \,\&\, (\Sigma Y_0)p. \tag{15}$$

Proof. For symmetry, apply Lemma 4 to the property $\lambda Y.\forall p(Y(\sim p) \leftrightarrow Yp)$. The base case follows from $K_\perp p = \perp$. Every successor is symmetric by (4). For the union condition, if

Bs, choose a symmetric stage $Y \subseteq B$ with Ys. Then $Y(\sim s)$, so $B(\sim s)$. Applying the same argument to $\sim s$ and using double negation gives the converse.

For the second assertion fix p and apply Lemma 4 to

$$\Psi = \lambda Y^u.\Big(Yp \to \exists Y_0^u(\mathrm{Stage}(Y_0) \,\&\, \sim Y_0 p \,\&\, (\Sigma Y_0)p)\Big).$$

The base case is vacuous. In the successor case suppose that $(\Sigma Y)p$. If Yp, the induction hypothesis supplies Y_0. If $\sim Yp$, take $Y_0 = Y$. For the union condition, an instance Bp has a stage $Y \subseteq B$ with Yp and $\Psi(Y)$. The latter supplies the desired Y_0. Generalizing p proves the assertion. ∎

Lemma 6. *The theory $\mathcal{T}$ proves that P is pure, P is a stage, and $L \subseteq P$.*

Proof. The term defining P uses the single nonlogical constant Pure_u. Let c have type $u \to t$, and let Cl_c be the closure formula with c replacing every occurrence of Pure_u. The term

$$M = \lambda c^{u\to t}.\lambda p.\exists Y^u\Big(cY \,\&\, \forall \mathcal{Z}^{u\to t}(\mathrm{Cl}_c(\mathcal{Z}) \to \mathcal{Z}(Y)) \,\&\, Yp\Big)$$

is closed and logical, of type $(u \to t) \to u$. By β-conversion, $M(\mathrm{Pure}_u) = P$. Logical purity gives $\mathrm{Pure}(M)$, while (PP_u) gives $\mathrm{Pure}_{u\to t}(\mathrm{Pure}_u)$. Application closure at input type $u \to t$ and output type u therefore gives $\mathrm{Pure}(P)$.

For each p with Pp, definition (13) supplies a stage Y with Yp, and $Y \subseteq P$ holds by the same definition. Since P is pure, the union condition for Stage applies to P and proves $\mathrm{Stage}(P)$. Successor closure gives $\mathrm{Stage}(\Sigma P)$, and hence

$$\Sigma P \subseteq P. \tag{16}$$

This is a closed theorem of the axiom extension $\mathcal{T}$. Its proof has used the Purity schema, Application closure, and (PP_u), with all temporary assumptions discharged. Necessitation therefore gives

$$\Box(\Sigma P \subseteq P).$$

Thus $\mathrm{Pre}(P)$ holds, by (2). Instantiating the quantifier in (3) at $W = P$ now gives $L \subseteq P$. ∎

PP supplies the inclusion $L \subseteq P$ used in the final argument. Lemma 3 supplies Lq, and Lemma 6 gives Pq. The definition of P then gives a stage containing q. Lemma 5 then supplies a pure Y_0 for which (15) holds at q.

Proof of Theorem 1. Suppose that r satisfies (R). Put $q = r \,\&\, Lr = gr$. By Lemmas 3 and 6, Pq. Choose a stage containing q. Lemma 5 gives a stage Y_0 with

$$\sim Y_0 q, \qquad (\Sigma Y_0) q.$$

It is pure and symmetric, so $\sim Y_0(\sim q)$ also holds. We consider the two disjuncts of $(\Sigma Y_0)q$.

Case 1. Suppose that

$$\Box(q \vee Y_0 q \vee Y_0(\sim q)).$$

Factivity gives the disjunction inside the box. Since its last two disjuncts are false, q holds. Define

$$X_+ = \lambda p.\big(gp \vee Y_0(gp) \vee Y_0(\sim(gp))\big).$$

Its purity follows from the Purity schema and Application closure, using the closed logical term

$$D_+ = \lambda Y^u.\lambda p.\big(gp \vee Y(gp) \vee Y(\sim(gp))\big)$$

applied to the pure Y_0. Here g expands to a closed logical term, so D_+ does too, with type $u \to u$. The identity $gr = q$ identifies $X_+ r$ with the entire disjunction in the box, giving $\Box(X_+ r)$. Condition (R) yields $\forall p\, X_+ p$, whose instance at $\sim q$ is $X_+(\sim q)$. Using $g(\sim q) = \sim q$ and double negation, we obtain

$$\sim q \vee Y_0(\sim q) \vee Y_0 q.$$

Each disjunct is false, which is a contradiction.

Case 2. Suppose that

$$\Box(\sim q \vee Y_0 q \vee Y_0(\sim q)).$$

Factivity and the same two local falsities give $\sim q$. Define

$$X_- = \lambda p.\big(\sim(gp) \vee Y_0(gp) \vee Y_0(\sim(gp))\big).$$

This is pure by logical purity of the corresponding term $D_- = \lambda Y^u.\lambda p.(\sim(gp) \vee Y(gp) \vee Y(\sim(gp)))$ and Application closure. Again $gr = q$ gives $\Box(X_- r)$. Condition (R) yields $\forall p\, X_- p$, and therefore $X_-(\sim q)$. Using $g(\sim q) = \sim q$ and double negation, we obtain

$$q \vee Y_0(\sim q) \vee Y_0 q.$$

All three disjuncts are false. This completes the contradiction in both cases, and proves that no r satisfies (R).

For the first consequence suppose that $\mathrm{fun}'(r)$ and that X is pure with $\Box(Xr)$. The operator $K_\top = \lambda p.\top$ is pure by the Purity schema. Since

$$Xr = \top = K_\top r,$$

the definition of $\mathrm{fun}'(r)$ yields $X = K_\top$. It follows that $Xp = \top$, hence Xp, for every p. Thus r satisfies (R), contradicting the result just proved.

Finally, unary Recombination states

$$\forall Xr\big((\mathrm{Pure}(X) \,\&\, \mathrm{Fun}(r)) \to (\Box(Xr) \to \forall p\, Xp)\big).$$

Every fundamental proposition would therefore satisfy (R). This proves the second consequence. ■